Assignment No.4

Magnetism in Condensed Matter, M. Phil Semester I, Session Fall 2025

Short Questions:

Q.1: In a large ferromagnetic crystal below the Curie temperature, the magnetization points along a specific direction.

a. Why is this direction not determined by the Hamiltonian?. 3 marks
b. What physical factors help select a particular direction?
c. Why is the symmetry breaking called *spontaneous*?

Q.2:

Explain why **true spontaneous symmetry breaking** cannot strictly occur in a finite magnetic system, but can occur in the thermodynamic limit. 3 marks

Q.3: By comparing the Ising and Heisenberg models of ferromagnetism, answer the following questions:

a. What symmetry is broken in each model? 3 marks
b. Why does the Ising model break a discrete symmetry?
c. How does symmetry affect the nature of excitations?

Q.4: An antiferromagnet has zero net magnetization in the ordered phase:

a. Does symmetry breaking still occur? 3 marks
b. What is the correct order parameter?
c. Which symmetry is broken?

Q.5 Near the Curie temperature, magnetization fluctuates strongly: 3 marks

a. Why do fluctuations increase near the phase transition?
b. How do these fluctuations affect symmetry breaking?
c. Why does mean-field theory fail near T_{C}?.

Q.1: In a large ferromagnetic crystal below the Curie temperature, the magnetization points along a specific direction.

a. Why is this direction not determined by the Hamiltonian?. 3 marks
b. What physical factors help select a particular direction?
c. Why is the symmetry breaking called *spontaneous*?

Ans:

Q.1 Ferromagnetism and spontaneous symmetry breaking

(a) Why is the direction not determined by the Hamiltonian?
Below the Curie temperature, the ferromagnetic Hamiltonian is *rotationally symmetric*: it has no preferred direction in space. All directions of uniform magnetization have the same energy. Hence the Hamiltonian fixes only the *magnitude* of the magnetization, not its direction. Choosing one direction among infinitely many equivalent ones is therefore not dictated by the Hamiltonian itself.

1(b) What physical factors help select a particular direction?
In real crystals, small perturbations break the perfect rotational symmetry and bias the choice of direction, such as:

- **Magnetocrystalline anisotropy** (spin–orbit coupling tied to the crystal lattice),
- **Dipolar (demagnetizing) fields**,
- **External magnetic fields**, even extremely weak ones,
- **Defects, strain, and boundaries** of the sample.
  These effects lift the degeneracy and favor specific "easy axes."

1(c) Why is the symmetry breaking called spontaneous?
It is called *spontaneous* because the ordered state (nonzero magnetization in a definite direction) has lower symmetry than the Hamiltonian, even in the absence of any explicit symmetry-breaking field. The system itself chooses one direction when cooled below the Curie temperature, despite all directions being energetically equivalent in the ideal Hamiltonian.

Q.2: Explain why true spontaneous symmetry breaking cannot strictly occur in a finite magnetic system, but can occur in the thermodynamic limit. 3 marks

Ans:

Q.2 Spontaneous symmetry breaking and system size

In a **finite magnetic system**, all symmetry-related states are connected by thermal or quantum fluctuations. Because the Hamiltonian is symmetric, the exact equilibrium state is a superposition or statistical mixture of all these equivalent orientations, giving **zero net magnetization** when averaged over time. Hence, *true* spontaneous symmetry breaking cannot strictly occur in a finite system.

In the **thermodynamic limit** (number of spins $N \to \infty$, volume $\to \infty$at fixed density), the energy barrier between different magnetization directions becomes infinite. Transitions between symmetry-related states are then suppressed, and the system can remain trapped in one ordered state indefinitely. As a result, a definite magnetization direction emerges and spontaneous symmetry breaking becomes well-defined.

Q.3: By comparing the Ising and Heisenberg models of ferromagnetism, answer the following questions:

a. What symmetry is broken in each model? 3 marks

b. Why does the Ising model break a discrete symmetry?

c. How does symmetry affect the nature of excitations?

Ans:

Q.3 Ising vs Heisenberg ferromagnets

3(a) What symmetry is broken in each model? (3 marks)

- **Ising model:** The Hamiltonian is invariant under a global spin flip $S_i \to -S_i$. Below the Curie temperature, the system chooses either "up" or "down" magnetization. Thus a **discrete** $\mathbb{Z}_2$symmetry is broken.
- **Heisenberg model:** The Hamiltonian is invariant under **continuous global spin rotations** (SO(3)). In the ordered phase, the magnetization selects a particular direction in spin space, so **continuous rotational symmetry** is broken.

3(b) Why does the Ising model break a discrete symmetry?
In the Ising model, each spin can point only in two directions (up or down). The only symmetry relating the two ordered states is a single global spin-flip operation. Since there is no continuum of equivalent directions, the symmetry being broken is **discrete**, not continuous.

3(c) How does symmetry affect the nature of excitations?

- **Ising model:** Breaking a discrete symmetry leads to **gapped excitations**, such as domain walls separating up- and down-magnetized regions.
- **Heisenberg model:** Breaking a continuous symmetry produces **gapless collective modes** (spin waves or magnons), which are the Goldstone modes associated with continuous symmetry breaking.

discrete symmetry → gapped excitations;

continuous symmetry → gapless Goldstone modes.

Q.4: An antiferromagnet has zero net magnetization in the ordered phase:

a. Does symmetry breaking still occur? 3 marks

b. What is the correct order parameter?

c. Which symmetry is broken?

Ans:

Q.4 Antiferromagnetism and symmetry breaking

4(a) Does symmetry breaking still occur? (3 marks)
Yes. Even though the **net magnetization is zero**, an antiferromagnet below the Néel temperature is in an *ordered* phase. The spins arrange themselves in a regular pattern (e.g. alternating up and down on two sublattices), which is not invariant under the full symmetry of the Hamiltonian. Hence, **spontaneous symmetry breaking does occur**.

4(b) What is the correct order parameter?
The correct order parameter is the **staggered magnetization** (or Néel vector),

$$\mathbf{L} = \frac{1}{N}\sum_{i}(-1)^{i}\mathbf{S}_{i},$$

which measures the difference between the magnetizations of the two sublattices. It is nonzero in the ordered phase even though the total magnetization is zero.

4(c) Which symmetry is broken?

- **Spin-rotation symmetry** (continuous, e.g. SO(3)) is broken when the Néel vector chooses a particular direction in spin space.
- In addition, **lattice translation symmetry by one lattice spacing** is broken, since exchanging the two sublattices changes the spin pattern.

Thus, antiferromagnets exhibit spontaneous breaking of both **spin rotational** and **discrete lattice** symmetries.

Q.5 Near the Curie temperature, magnetization fluctuates strongly: 3 marks

a. Why do fluctuations increase near the phase transition?

b. How do these fluctuations affect symmetry breaking?

c. Why does mean-field theory fail near TC ?.

Ans:

Q.5 Critical fluctuations near the Curie temperature

5(a) Why do fluctuations increase near the phase transition? (3 marks)
Near the Curie temperature T_C, the free-energy cost of changing the magnetization becomes very small. As a result, **large regions of spins become correlated**, and the correlation length grows (formally diverges at T_C). Thermal energy can then easily produce large fluctuations of magnetization over many length scales.

5(b) How do these fluctuations affect symmetry breaking?
Strong fluctuations continually restore the symmetry on large length and time scales. Exactly at T_C, long-range order is destroyed and the order parameter averages to zero, so **symmetry is not broken**. Only below T_C, when fluctuations are sufficiently suppressed, can a stable symmetry-broken state with nonzero magnetization exist.

5(c) Why does mean-field theory fail near T_C?
Mean-field theory assumes that each spin feels only an *average* field and **neglects fluctuations and spatial correlations**. Near T_C, fluctuations occur on all length scales and dominate the physics, invalidating this assumption. Consequently, mean-field theory gives incorrect critical behavior (wrong critical exponents) close to the Curie temperature.